

Full Data Science Roadmap (Beginner to Advanced)

This roadmap is designed to guide you step-by-step through learning data science, from fundamentals to advanced projects. Start by following each phase in sequence, mastering the core concepts before moving ahead. For every topic, you'll find carefully selected free resources—mostly hands-on tutorials and practical projects—to help you learn by doing. Take your time to practice, explore different datasets, and build your portfolio as you progress. Later, we will break down each topic further with detailed, free resources and brief explanations to help you choose the best learning path for your needs.

Phase 1: Prerequisites & Foundation

1. Mathematics for Data Science

- Linear Algebra
- Probability
- Statistics (Descriptive & Inferential)
- Calculus (basic level)

2. Programming with Python

- Syntax & Logic
- Functions & Loops
- File Handling
- OOPs Concepts
- Libraries: NumPy, Pandas

Phase 2: Data Handling & Analysis

3. Data Analysis & Manipulation

- Pandas for DataFrames

- NumPy for numerical ops
- Data Cleaning & Wrangling
- Working with Real Datasets (CSV, Excel, JSON)

4. Data Visualization

- Matplotlib & Seaborn
- Plotly & Dash (Optional)
- Storytelling with Charts
- Visualizing Correlations and Trends

Phase 3: Machine Learning Core

5. Supervised Learning

- Linear Regression
- Logistic Regression
- Decision Trees & Random Forest
- SVM, KNN

6. Unsupervised Learning

- K-Means Clustering
- Hierarchical Clustering
- PCA (Dimensionality Reduction)

7. Model Evaluation & Tuning

- Train/Test Split
- Cross-Validation

- Confusion Matrix, Accuracy, Precision/Recall
- Hyperparameter Tuning (GridSearch, RandomSearch)

Phase 4: Advanced Concepts

8. Feature Engineering & Selection

- Handling Missing Data
- Encoding & Scaling
- Feature Importance
- Domain-Based Features

9. Time Series Analysis

- Stationarity
- ARIMA/SARIMA
- Forecasting Models
- Real-world datasets (stock, weather)

10. Introduction to Deep Learning

- Basics of Neural Networks
- Perceptrons
- Forward/Backward Propagation
- Keras/TensorFlow Basics

Phase 5: Real-World Applications & Deployment

11. Project Building & Capstone Projects

- Regression: House Price Prediction

- Classification: Titanic Survival, Loan Default
- NLP: Sentiment Analysis
- Clustering: Customer Segmentation

12. Model Deployment (MLOps Basics)

- Streamlit/Flask APIs
- Docker Basics
- Deploy to Hugging Face/Render/Heroku
- Git & GitHub for Project Management

Phase 6: Domain Integration & Extras

13. SQL for Data Science

- SELECT, JOIN, GROUP BY
- Subqueries
- Window Functions
- Hands-on with public datasets

14. Cloud Tools for DS (Optional Advanced)

- Google Colab
- BigQuery Basics
- AWS S3, SageMaker intro

Now that you have an overview of the complete data science roadmap, let's break down each phase and concept into more detailed sections.

This will include step-by-step learning paths, free hands-on resources, and brief explanations of why each resource is valuable. This detailed breakdown will help you focus on one topic at a time and make your learning journey clear and manageable.

Phase 1: Mathematics for Data Science

Topics Covered:

1. **Linear Algebra**
2. **Probability**
3. **Statistics**
4. **Basic Calculus (optional but helpful)**

1. Linear Algebra

Resource 1: [Essence of Linear Algebra – 3Blue1Brown \(YouTube\)](#)

- Platform: YouTube – 3Blue1Brown
- This visual playlist is *famous* for making abstract linear algebra intuitive through animations. Covers vectors, matrices, linear transformations, etc.
- **Why Learn:** This builds your **geometric understanding** of matrix operations, a must for understanding **machine learning models** and PCA.

Resource 2: [Linear Algebra for Data Science – freeCodeCamp \(YouTube\)](#)

- Platform: YouTube – freeCodeCamp
- A long-form practical course for data science learners. Focuses on matrix math with Python, NumPy operations, and real use cases.
- **Why Learn:** Great for **hands-on learners** who want to mix math with coding from Day 1.

Resource 3: [Khan Academy – Linear Algebra](#)

- Platform: Khan Academy
- Structured lessons with quizzes, examples, and visual explanations. Very beginner-friendly, focused on theory with occasional visuals.
- **Why Learn:** Best for those who prefer **step-by-step foundations**, and it's easy to revisit topics at your own pace.

2. Probability

Resource 1: [Probability Fundamentals – Khan Academy \(YouTube\)](#)

- Platform: YouTube – Khan Academy
- Simple explanations with cartoons. It breaks down difficult probability terms like independence, Bayes' Theorem, distributions, etc.
- **Why Learn:** Extremely clear if you're scared of math. Great to **build confidence in probability thinking**.

Resource 2: [Introduction to Probability – Harvard \(EdX Audit\)](#)

- Platform: HarvardX (Free via self-study site)
- A rigorous but approachable course built by Harvard for undergrads, now made free. Includes PDFs, lecture notes, and problem sets.
- **Why Learn:** Industry-level depth. Perfect if you want to go **beyond basic intuition and into theory**.

Resource 3: [Khan Academy – Probability & Combinatorics](#)

- Platform: Khan Academy
- Bite-sized videos + hands-on quizzes. Explains coin flips, dice, conditional probability, and combinations.
- **Why Learn:** Ideal if you're starting from scratch and want a **structured, gentle path**.

3. Statistics (Descriptive + Inferential)

Resource 1: [CrashCourse – Statistics \(YouTube\)](#)

- Platform: YouTube – CrashCourse
- Fast-paced, visual and very engaging! Covers mean, median, variance, distribution types, z-scores, etc.
- **Why Learn:** Great for **quick foundational review** and building familiarity with statistical language.

Resource 2: [Statistics for Data Science – freeCodeCamp \(YouTube\)](#)

- Platform: YouTube – freeCodeCamp
- A full 3-hour stats course specifically geared toward **real-world data science applications**. Includes practical implementation with Python.
- **Why Learn:** Combines theory + coding, ideal for project-based learners who want immediate application.

4. Basic Calculus (Optional but Useful)

Resource 1: [Calculus for Machine Learning – YouTube](#)

- Platform: YouTube – Jon Krohn
- Shows how derivatives and gradients apply in neural networks and optimization.
- **Why Learn:** If you plan to learn **deep learning**, this will help you understand how models *actually learn*.

Resource 2: [Khan Academy – Differential Calculus](#)

- Platform: Khan Academy
- A full series starting from “What is a derivative?” to applications like slope, rate of change, etc.
- **Why Learn:** Best for **zero background** students — self-paced with visuals and quizzes.

Phase 2: Programming with Python for Data Science

Topics Covered:

1. Core Python Fundamentals
2. Python for Data Handling (NumPy & Pandas)
3. Hands-on Coding Practice

1. Core Python Fundamentals

Resource 1: [Python for Beginners – Programming with Mosh \(YouTube\)](#)

- Platform: YouTube – Programming with Mosh
- Covers variables, loops, functions, error handling, and OOP—all beginner-friendly.
- Why Learn: Mosh is known for **clear, no-fluff teaching** — this video (6+ hours) gives a **solid Python base in one sitting**.

Resource 2: [Python Basics – freeCodeCamp \(YouTube\)](#)

- Platform: YouTube – freeCodeCamp
- Starts from zero: covers syntax, loops, functions, OOP, etc., and includes small exercises throughout.
- Why Learn: Designed for **complete beginners**, and the examples are great for **logic building**.

Resource 3: [Python Practice – W3Schools \(Interactive\)](#)

- Platform: W3Schools
- Click-to-run Python code with built-in exercises for each topic (functions, loops, strings, etc.)
- Why Learn: Useful for **hands-on practice instantly** without needing to install anything.

2. Python for Data Handling (NumPy & Pandas)

Resource 1: [NumPy Tutorial – freeCodeCamp \(YouTube\)](#)

- Platform: YouTube – freeCodeCamp
- A full practical tutorial teaching array creation, indexing, broadcasting, and mathematical ops using NumPy.
- Why Learn: NumPy is the **foundation for all numerical computing in ML/AI** — this makes you job-ready in handling arrays.

Resource 2: [Pandas Tutorial – Data School \(YouTube Playlist\)](#)

- Platform: YouTube – Data School
- Short videos explaining concepts like Series, DataFrames, filtering, grouping, merging, etc.
- Why Learn: Covers **real-world use cases** in short segments—perfect for revision and project prep.

Resource 3: [Pandas Official Tutorials – pandas.pydata.org](#)

- Platform: Pandas Official Site
- Authoritative documentation with code examples, ideal for deep diving into any feature.
- Why Learn: Knowing how to **read official docs** is a long-term skill for any serious coder/data scientist.

3. Hands-on Coding Practice

Resource 1: [HackerRank – Python Challenges](#)

- Platform: HackerRank
- 10 Days of Python Practice – great for data science beginners to sharpen their problem-solving.
- Why Learn: Ideal for **developing logic-building and syntax recall**—vital in interviews.

Resource 2: [Kaggle – Python Course \(Free\)](#)

- Platform: Kaggle
- Interactive notebooks focused on writing and running Python code in the browser. Focused on **data-centric tasks**.
- Why Learn: It's *hands-on, beginner-friendly*, and **directly aligned to data science practice**.

Resource 3: [Exercism – Python Track](#)

- Platform: Exercism
- Offers mentor-supported Python problems. Step-by-step and challenges that go deeper than syntax.
- Why Learn: Encourages **code quality and thought process** — useful once you've mastered basics.

Phase 3: Data Analysis & Visualization

Topics Covered:

1. Data Cleaning & Exploration with Pandas
2. Data Visualization with Matplotlib, Seaborn & Plotly
3. Working with Real Datasets (CSV, Excel, APIs)

1. Data Cleaning & Exploration with Pandas

Resource 1: [Exploratory Data Analysis – Kaggle \(Free\)](#)

- Platform: Kaggle
- Hands-on notebook lessons covering missing data, string handling, type conversion, and common EDA workflows.

- Why Learn: Helps you **think like a data analyst** — not just clean data, but *understand what's wrong with it*.

Resource 2: [Data Analysis with Python – freeCodeCamp \(YouTube\)](#)

- Platform: YouTube – freeCodeCamp
- Covers reading datasets, grouping, filtering, applying functions, and summarizing data. Full project-based teaching.
- Why Learn: This **mimics a real analysis scenario**, good for beginners starting their first mini-projects.

Resource 3: [Real-Time EDA on Titanic Dataset – Krish Naik \(YouTube\)](#)

- Platform: YouTube – Krish Naik
- Step-by-step analysis on the famous Titanic dataset. Shows how to draw meaning from columns and relations.
- Why Learn: Gives you **real project experience** and teaches how to build a storytelling mindset.

2. Data Visualization (Matplotlib, Seaborn, Plotly)

Resource 1: [Matplotlib & Seaborn Crash Course – freeCodeCamp \(YouTube\)](#)

- Platform: YouTube – freeCodeCamp
- Full 2-hour course showing how to plot bar charts, scatter plots, heatmaps, histograms with real datasets.
- Why Learn: Helps you **build dashboards and reports** — a vital part of analyst and DS roles.

Resource 2: [Interactive Plots with Plotly – Harry's data Journey \(YouTube\)](#)

- Platform: YouTube – Harry's Data Journey

- Introduces Plotly to create interactive plots, charts, and dashboards directly from notebooks.
- Why Learn: Adds **interactivity to your portfolio**, especially useful in apps or dashboards.

3. Working with Real Datasets

Resource 1: [Kaggle Datasets – Start Exploring](#)

- Platform: Kaggle
- A library of 50,000+ real-world datasets across finance, health, education, games, etc.
- Why Learn: Practicing with real data builds **project-ready confidence** and problem-solving skills.

Resource 2: [How to Import Data \(CSV, Excel, APIs\) – CodeBasics \(YouTube\)](#)

- Platform: YouTube – CodeBasics
- Shows how to read CSVs, connect to APIs, extract Excel files, and clean data automatically.
- Why Learn: Understanding **multiple data sources** prepares you for enterprise-level projects.

Resource 3: [Real-World Projects – YouTube \(Codebasics\)](#)

- Platform: YouTube – CodeBasics
- Hands-on projects like IPL data analysis, YouTube revenue prediction, and customer segmentation.
- Why Learn: Helps you **practice storytelling and presentation**, which are core in real interviews.

Phase 4: Machine Learning (ML)

This is where your data starts turning into **predictions** and **insights that drive action**. You'll explore how machines learn patterns, make decisions, and how you can build models that generalize well to new data.

Topics Covered:

1. Introduction to Machine Learning
2. Supervised Learning Algorithms
3. Unsupervised Learning Algorithms
4. Model Evaluation & Tuning
5. ML Projects (Hands-on)

1. Introduction to Machine Learning

Resource 1: [Machine Learning for Beginners – freeCodeCamp \(YouTube\)](#)

- Platform: YouTube – freeCodeCamp
- A 4-hour beginner course by Simplilearn that starts from scratch and covers core ML concepts.
- Why Learn: You'll understand **what ML is**, types of learning (supervised/unsupervised), model building pipeline, and basic terms like underfitting, overfitting, bias, and variance.

Resource 2: [Kaggle: Intro to Machine Learning \(Free\)](#)

- Platform: Kaggle
- Code-along mini-course teaching how to predict house prices using Decision Trees and Random Forests.
- Why Learn: Helps **apply ML hands-on** even if you're new. Easy-to-understand concepts + real dataset usage.

Resource 3: [ML Basics – StatQuest \(YouTube Playlist\)](#)

- Platform: YouTube – StatQuest

- Short, animated explainers that break down ML math intuitively (e.g., linear regression, trees).
- Why Learn: StatQuest is **gold for conceptual clarity** with fun, cartoon-style teaching.

2. Supervised Learning Algorithms

Resource 1: [Supervised ML with Scikit-learn – freeCodeCamp \(YouTube\)](#)

- Platform: YouTube – freeCodeCamp
- Teaches classification, regression using Python's Scikit-learn with projects like Iris classification.
- Why Learn: Shows how to write ML pipelines with real data, train/test split, and model interpretation.

Resource 2: [Kaggle: Intermediate ML \(Handling Missing Values, Categorical Data\)](#)

- Platform: Kaggle
- Expands on ML intro by dealing with real-world mess: missing values, categorical features, leakage.
- Why Learn: Takes you one level deeper to **write professional-grade ML code**.

Resource 3: [ML Algorithms – Krish Naik Playlist \(YouTube\)](#)

- Platform: YouTube – Krish Naik
- Practical explanation of Linear Regression, Logistic Regression, SVM, KNN, etc., with full implementation.
- Why Learn: Builds **confidence in coding ML algorithms** end-to-end, not just importing them.

3. Unsupervised Learning Algorithms

Resource 1: [Unsupervised ML – freeCodeCamp \(YouTube\)](#)

- Platform: YouTube – freeCodeCamp
- Explains Clustering, K-Means, PCA, and Dimensionality Reduction with Python.
- Why Learn: Essential for **segmentation tasks, recommendation systems**, and reducing large datasets.

Resource 2: [KMeans Clustering – StatQuest \(YouTube\)](#)

- Platform: YouTube – StatQuest
- Animated breakdown of how KMeans algorithm works.
- Why Learn: Understand how unsupervised learning **really works** without jumping into code first.

4. Model Evaluation & Tuning

Resource 1: [Model Evaluation Metrics – StatQuest \(YouTube\)](#)

- Platform: YouTube – StatQuest
- Covers accuracy, precision, recall, confusion matrix, ROC curve, etc.
- Why Learn: These metrics are **asked in every interview** — crucial to justify your model's performance.

Resource 2: [GridSearchCV](#)

[CorssValidation](#)

- Platform: YouTube – 2 channels codebasics and stackQuest
- Step-by-step guide to hyperparameter tuning using GridSearchCV and RandomizedSearchCV.
- Why Learn: Tuning models **improves performance drastically**, and this video teaches it cleanly.

5. ML Projects (Hands-On)

Resource 1: [Titanic ML Project – Kaggle Competition](#)

- Platform: Kaggle
- Classic beginner-friendly classification project with starter notebooks and forums.
- Why Learn: First real ML project that teaches **data prep, model training, and evaluation**.

Resource 2: [End-to-End Projects – CodeBasics \(YouTube\)](#)

- Platform: YouTube – CodeBasics
- Real-world projects (house price prediction, loan default prediction, HR analytics).
- Why Learn: Great for building **portfolio-worthy case studies** using structured industry problems.

Phase 4.5: Deep Learning

What You'll Learn:

Master the art of building models that can learn from **images, text, and sequences**. Deep learning enables everything from **image recognition** to **language translation** and **chatbots**.

1. Neural Network Fundamentals

Resource 1: [Neural Networks & Deep Learning – 3Blue1Brown](#)

Platform: YouTube

Visually intuitive series explaining how neurons, weights, and backpropagation work using animations.

Why Learn: Makes abstract math like gradients and layers very simple with brilliant animations. Great starting point.

Resource 2: [Deep Learning Specialization – Andrew Ng \(Audit Free\)](#)

Platform: Coursera

One of the most trusted beginner series covering NN architectures, loss functions, and forward/backward pass.

Why Learn: Combines foundational math with hands-on coding exercises. Industry-relevant content.

Resource 3: [Neural Networks from Scratch – Sentdex](#)

Platform: YouTube

Build a neural net from scratch in Python & NumPy — no libraries, just logic.

Why Learn: If you love learning the hard way, this gives true clarity on what's happening

under the hood.

Practice Idea:

- Build a neural net from scratch to solve XOR problem or digit classification
- Visualize activation values layer by layer

2. Deep Learning Frameworks (TensorFlow & PyTorch)

Resource 1: [TensorFlow in 100 Minutes – freeCodeCamp](#)

Platform: YouTube

Crash course on TensorFlow & Keras with a small image classifier project.

Why Learn: Best intro if you want to go from zero to working model quickly.

Resource 2: [PyTorch Tutorial for Beginners – Aladdin Persson](#)

Platform: YouTube

Learn tensors, gradients, and building networks with PyTorch.

Why Learn: PyTorch is widely used in research. This helps you switch between frameworks easily.

Resource 3: [Official Keras Documentation](#)

Platform: keras.io

Tutorials for building models — regression, classification, image, etc.

Why Learn: Clean examples and official guidance for beginners using Keras.

Practice Idea:

- Build a regression model in Keras (predict house prices)
- Rebuild the same in PyTorch

3. Convolutional Neural Networks (CNNs)

Resource 1: [CNNs Explained – StatQuest](#)

Platform: YouTube

Simple animated explanation of how filters, kernels, pooling layers work.

Why Learn: Turns complex CNN concepts into fun stories with visuals.

Resource 2: [Deep Learning for Computer Vision – freeCodeCamp](#)

Platform: YouTube

Build CNNs using TensorFlow on real datasets like MNIST.

Why Learn: Combines code + theory in one long-form tutorial.

Resource 3: [Intro to Deep Learning \(CNNs\) – Kaggle](#)

Platform: Kaggle

Interactive notebook with CNN hands-on practice on MNIST.

Why Learn: Directly practice in-browser, no setup needed.

Practice Idea:

- Build a CNN to classify fashion images (Fashion MNIST)
- Try visualizing feature maps and learned filters

4. RNNs & LSTMs (For Sequences and Text)

Resource 1: [RNNs and LSTMs – StatQuest](#)

Platform: YouTube

Explains how RNNs and LSTMs work with sequences.

Why Learn: Makes time-based learning (stocks, text) easy to understand.

Resource 2: [NLP with Deep Learning – freeCodeCamp](#)

Platform: YouTube

Sentiment analysis using LSTMs in Keras.

Why Learn: Combines NLP theory with model training and evaluation.

Resource 3: [PyTorch NLP Tutorials](#)

Platform: pytorch.org

Step-by-step text generation and classification using PyTorch.

Why Learn: Use real PyTorch code for NLP pipelines.

Practice Idea:

- Predict next word or generate text (e.g., Shakespeare-style text)
- Train an LSTM model to forecast stock or temperature

5. Transfer Learning

Resource 1: [Hugging Face Transformers Crash Course](#)

Platform: YouTube

Intro to transformer-based NLP models like BERT.

Why Learn: Hugging Face is the standard in NLP — must-have for language projects.

Resource 2: [Kaggle Transfer Learning – Cats vs Dogs](#)

Platform: Kaggle

Fine-tune a pre-trained CNN to classify cat vs dog images.

Why Learn: Real-world practice with minimal data using pre-trained models.

Practice Idea:

- Fine-tune ResNet for custom flower image classification
- Use BERT for text classification (positive/negative reviews)

6. Hands-on Deep Learning Projects

Resource 1: [CNN Image Classification – Kaggle](#)

Platform: Kaggle

Step-by-step project using CNNs to classify dog vs cat images.

Why Learn: Combines preprocessing, model building, training, and submission.

Resource 2: [Sentiment Analysis with LSTM – Towards Data Science](#)

Platform: Medium

Uses LSTM model to analyze text sentiment.

Why Learn: Great starter NLP project with deep learning integration.

Resource 3: [Time Series Forecasting with LSTMs – Krish Naik](#)

Platform: YouTube

Forecast stock prices or similar time series using deep learning.

Why Learn: Real-world sequence modeling in a simple, clear format.

Search: “Krish Naik LSTM Time Series”

Practice Idea:

- Build a Streamlit app to classify X-rays or flowers
- Deploy a Hugging Face transformer for tweet sentiment
- Forecast electricity usage with LSTMs

Phase 5: SQL & Databases for Data Science

SQL is *non-negotiable* for data scientists. Whether you're pulling customer data, joining transactional logs, or filtering millions of rows — SQL is the language of data access. It's a must-have for both interviews and real jobs.

Topics Covered:

1. SQL Basics: SELECT, WHERE, ORDER BY
2. Aggregations: GROUP BY, COUNT, SUM, AVG
3. Joins: INNER, LEFT, RIGHT, FULL
4. Subqueries, Window Functions, CTEs
5. Hands-on SQL Projects

1. SQL Basics

Resource 1: [SQL for Data Science – freeCodeCamp \(YouTube\)](#)

- Platform: YouTube – freeCodeCamp
- Covers SQL syntax, filtering, sorting, aliasing, and basic aggregation using MySQL.
- Why Learn: This is your **go-to crash course**, taught like a bootcamp from zero to real-world usage.

Resource 2: [SQL Bolt – Interactive Lessons](#)

- Platform: SQLBolt
- Click-and-run interactive tutorials — from basic SELECTs to JOINS — with explanations and practice.
- Why Learn: Great for **absolute beginners** who want to learn by doing in-browser without setup.

Resource 3: [Mode SQL Tutorial – Business Focused](#)

- Platform: Mode Analytics
- Lessons on querying real business datasets. Explains not just syntax, but **why** you write a query.
- Why Learn: Helps you **think like an analyst**, writing SQL for product, business, or growth teams.

2. Aggregations & Filtering

Resource 1: [LeetCode SQL Easy–Medium Questions](#)

- Platform: LeetCode
- Hands-on problems that test your use of COUNT, GROUP BY, and filtering with real-world context.
- Why Learn: Good for **interview-style questions** and building muscle memory in query writing.

Resource 2: [StrataScratch SQL Practice](#)

- Platform: StrataScratch
- Real company datasets (Google, Amazon, Uber) and tasks like top customers, avg spend, churn prediction.
- Why Learn: You learn **how SQL is used in the wild** — perfect for preparing for job assessments.

3. Joins, Subqueries & Window Functions

Resource 1: [Joins Explained Visually – Khan Academy](#)

- Platform: Khan Academy
- Visual diagrams explaining INNER, LEFT, RIGHT, FULL joins and how they work.
- Why Learn: Makes **join logic crystal clear**, especially for visual learners.

Resource 2: [Window Functions & CTEs – Mode SQL Advanced](#)

- Platform: Mode Analytics
- Learn how to use advanced SQL to rank users, calculate moving averages, and segment customers.
- Why Learn: These tools are **used in advanced analytics** and product metrics — boosts your skillset fast.

4. Hands-on SQL Projects

Resource 1: [Kaggle: SQL Datasets \(Google BigQuery\)](#)

- Platform: Kaggle
- Practice SQL queries on datasets like Stack Overflow, US Census, and Google Analytics using BigQuery.
- Why Learn: Write **live queries** on massive real datasets for portfolio building.

Resource 2: [8 SQL Case Studies – DataLemur](#)

- Platform: DataLemur
- Real-world company questions turned into SQL challenges: average order value, active user count, etc.
- Why Learn: Excellent **mock interview practice** with solutions and walkthroughs.

Phase 6: Statistics & Probability for Data Science

Statistics is the **foundation** of Data Science. Without it, you're just plotting numbers. It helps you **analyze data**, **validate models**, and **draw actionable conclusions** from experiments or trends.

Topics Covered:

1. Descriptive Statistics
2. Probability Basics

3. Bayes' Theorem & Conditional Probability
4. Distributions (Normal, Binomial, etc.)
5. Hypothesis Testing
6. Hands-on Stats Projects

1. Descriptive Statistics

Resource 1: [Statistics for Data Science – freeCodeCamp \(YouTube\)](#)

- Platform: YouTube – freeCodeCamp
- Full course on mean, median, mode, variance, standard deviation, data spread, and distributions.
- Why Learn: Covers **real examples** to explain concepts in the context of data science and analytics.

Resource 2: [Descriptive Stats – Khan Academy](#)

- Platform: Khan Academy
- Bite-sized videos and quizzes on central tendency, box plots, and histograms.
- Why Learn: **Super beginner-friendly** with animations and interactive practice.

Resource 3: [Basic Stats Using Python – W3Schools](#)

- Platform: W3Schools
- Teaches mean, median, mode, and std dev using Python step-by-step.
- Why Learn: Bridges stats theory with **Python code instantly** — great for applied learners.

2. Probability & Distributions

Resource 1: [Probability Basics – 3Blue1Brown \(YouTube\)](#)

- Platform: YouTube – 3Blue1Brown

- Visual explanation of probability, combinations, permutations, and random variables.
- Why Learn: Stunning **intuitive animations** for understanding “why probability works.”

Resource 2: [StatQuest Probability + Distributions](#)

- Platform: YouTube – Zedstatictics
- Covers normal, binomial, Poisson, and uniform distributions clearly and with fun.
- Why Learn: Josh makes **scary concepts fun and friendly**. Perfect for foundational clarity.

Resource 3: [Probability in Python – PythonLikeYouMeanIt](#)

- Platform: Data camp
- Shows how to simulate and visualize probabilities in Python using NumPy.
- Why Learn: Teaches **how to code probabilities** — necessary for simulations, ML, and stats modeling.

3. Hypothesis Testing & Confidence Intervals

Resource 1: [Hypothesis Testing – Codebasics \(YouTube\)](#)

- Platform: YouTube – Codebasics
- Explains p-value, z-test, t-test using Excel and Python.
- Why Learn: It's **project-based and practical**, with relatable business examples.

Resource 2: [Hypothesis Testing – StatQuest](#)

- Platform: YouTube – StatQuest
- Crisp and cartoon-style breakdown of null/alternate hypotheses, Type I/II errors.
- Why Learn: Makes statistical testing **easy to visualize and remember**.

Resource 3: [T-Tests & Confidence Intervals – Medium\(Blog\)](#)

- Platform: Medium
- Text-based, clean guide to hypothesis testing with Python examples.
- Why Learn: Helps understand how **real data scientists apply testing** in notebooks.

4. Hands-on Statistics Practice

Resource 1: [Kaggle: Statistics Course](#)

- Platform: Kaggle
- Teaches descriptive stats, sampling, distributions, and statistical testing.
- Why Learn: Code + concept combo with **real datasets** and in-browser exercises.

Resource 2: [Mini Statistics Projects – DataCamp \(Free Chapters\)](#)

- Platform: DataCamp (first chapters free)
- Uses Python to explore datasets with confidence intervals and histograms.
- Why Learn: Project-oriented and **Pythonic application** of stats.

Phase 7: Data Visualization

Once you've gathered insights from your data, **how you present it matters**. Data visualization helps you *tell stories*, *explain trends*, and *support decisions*. Good visualizations can speak louder than tables of numbers.

Topics Covered:

1. Matplotlib – The Foundation
2. Seaborn – Advanced Statistical Plots
3. Plotly – Interactive Dashboards
4. Tableau & Power BI (Free Tools)

5. Data Storytelling & Real-World Dashboards

1. Matplotlib – The Foundation of Python Visualizations

Resource 1: [Matplotlib Crash Course – Corey Schafer \(YouTube\)](#)

- Platform: YouTube – Corey Schafer
- Teaches the fundamentals: line plots, bar charts, labels, legends, saving figures.
- Why Learn: Matplotlib is the **core library** for all Python plots — essential for any visualization work.

Resource 2: [W3Schools Matplotlib](#)

- Platform: W3Schools
- Simple and clear tutorial with live coding blocks.
- Why Learn: Great for **hands-on trial and error** without setting up Jupyter.

Resource 3: [Official Matplotlib Tutorials](#)

- Platform: Matplotlib.org
- Detailed documentation with examples and use cases.
- Why Learn: Go here when you're ready to **customize and master** Matplotlib deeply.

2. Seaborn – Statistical Plots Made Easy

Resource 1: [Seaborn Tutorial – KGP talkie \(YouTube\)](#)

- Platform: YouTube – KGP talkie
- Covers line plots, scatter plots, boxplots, histograms, heatmaps — all with real datasets.
- Why Learn: Seaborn builds on Matplotlib, making **beautiful plots with minimal code**.

Resource 2: [Seaborn Documentation + Gallery](#)

- Platform: Seaborn
- Visual gallery with code for each plot type.
- Why Learn: Perfect for **inspiration and copying working code** when building dashboards.

Resource 3: [Kaggle: Data Visualization Course](#)

- Platform: Kaggle
- Covers Seaborn + pandas viz with hands-on code challenges.
- Why Learn: You can **code and submit instantly** inside the browser — no setup needed.

3. Plotly – For Interactive Visualizations

Resource 1: [Plotly Crash Course – Tech with Tim \(YouTube\)](#)

- Platform: YouTube – Tech With Tim
- Shows bar plots, line plots, interactive tooltips, and web-based dashboards.
- Why Learn: Plotly lets you create **interactive, zoomable charts** easily with Python.

Resource 2: [Official Plotly Docs \(Python\)](#)

- Platform: Plotly
- Step-by-step tutorials for each chart: area, scatter, 3D, map plots, etc.
- Why Learn: Master Plotly **directly from the source** with working notebooks.

Resource 3: [Build Dashboards with Plotly Dash – YouTube](#)

- Platform: YouTube – Charming Data
- Full tutorial on using Dash to build dashboards and web apps using Plotly charts.
- Why Learn: Lets you **build live data dashboards** for your resume and projects.

4. Power BI & Tableau (Free Options)

Resource 1: [Power BI Full Course – Avi Singh powerBIPor \(YouTube\)](#)

- Platform: YouTube – AVI singh PowerBIPor
- Learn Power BI basics, DAX, data modeling, and visualizations from scratch.
- Why Learn: Power BI is a **top business intelligence tool** used by companies across industries.

Resource 2: [Tableau Public + Full Tutorial – Tableau Tim \(YouTube\)](#)

- Platform: YouTube – Tableau Tim
- Learn how to use Tableau Public (free version) to create charts, dashboards, and storyboards.
- Why Learn: Tableau is great for **drag-and-drop storytelling**, useful for non-coders too.

Resource 3: [Google Data Studio \(Looker Studio\) + Tutorial](#)

- Platform: Google
- Build live dashboards with Google Sheets or BigQuery for free.
- Why Learn: **Cloud-powered dashboards** with zero code — ideal for portfolio work and reporting.

5. Hands-on Visualization Projects

Resource 1: [Kaggle Visualization Projects](#)

- Platform: Kaggle
- Explore 1,000s of public notebooks analyzing Titanic, COVID-19, sales data, etc.
- Why Learn: Hands-on projects help you **learn by doing**, and you can fork others' work.

Resource 2: [Makeover Monday Dataset + Tableau Challenges](#)

- Platform: Makeover Monday
- Weekly dataset challenges with a dashboard design theme.
- Why Learn: Improve **data storytelling skills** and post weekly on LinkedIn for visibility.

Resource 3: [Power BI Sample Dashboards – Microsoft](#)

- Platform: Microsoft Docs
- Downloadable datasets for retail, finance, HR, and marketing to practice dashboards.
- Why Learn: Practice **real-world scenarios** with enterprise-ready datasets.

Phase 8: End-to-End Data Science Projects (Mini to Major)

Now it's time to **apply everything you've learned** — data cleaning, exploration, statistics, visualization, and Python — into real-world projects.

Doing projects helps you:

- Build a **portfolio** for your resume or GitHub.
- Learn how data flows from **raw to insights**.
- Prepare for interviews (most DS interviews ask for project explanations).

What You'll Do in This Phase:

1. Mini Projects (Basic to Intermediate)
2. Major Projects (End-to-End with Models)
3. Kaggle Competitions / Hackathons
4. Portfolio Building & Documentation

1. Mini Projects (Perfect for Beginners)

Resource 1: [Kaggle Titanic Survival Prediction](#)

- Platform: Kaggle
- Classic dataset where you clean data, handle missing values, and build a basic ML model.
- Why Learn: It's the **"hello world"** of data science — great intro to structured projects.

Resource 2: [Data Cleaning & EDA with Netflix Dataset – GitHub](#)

- Platform: GitHub
- Use pandas, seaborn, and matplotlib to clean and visualize Netflix content data.
- Why Learn: Teaches real-life **exploratory data analysis** with plots and insights.

2. Major Projects (Intermediate to Advanced)

Resource 1: [Retail Sales Forecasting – GitHub Project](#)

- Platform: YouTube – GitHub
- Full series with Excel → Python → ML → dashboard reporting.
- Why Learn: Gives a **realistic end-to-end business project feel** with KPIs.

Resource 2: [HR Analytics & Attrition Prediction – GitHub Project](#)

- Platform: GitHub
- Use classification models to predict which employees might leave.
- Why Learn: It covers **feature engineering + classification** on real HR data.

Resource 3: [Zomato Data Analysis & Visualization – YouTube](#)

- Platform: YouTube – The iScale

- Complete analysis + dashboard on Zomato restaurant data using pandas & matplotlib.
- Why Learn: You learn **data storytelling + dashboard building** on real urban dataset.

3. Kaggle Competitions + Datasets

Resource 1: [Kaggle Competitions Page](#)

- Platform: Kaggle
- Join live challenges like credit scoring, fraud detection, sales predictions.
- Why Learn: Lets you **compete, learn from others, and get ranked** in the community.

Resource 2: [Awesome Public Datasets – GitHub](#)

- Platform: GitHub
- 100s of datasets across domains like finance, sports, agriculture, and education.
- Why Learn: Find **unique datasets** for personalized projects and ideas.

Resource 3: [Data Science Hackathons – Analytics Vidhya](#)

- Platform: Analytics Vidhya
- Timed hackathons and long-term ML contests with cash prizes.
- Why Learn: Simulates **real-world competition-style thinking**, great for practice.

- With ❤️ Harsha Verse